

Enterprise Data Cleaning & Quality Assurance Report

Organization: WeIntern Pvt Ltd | Author: Bineeta Yadav (Data Science Intern) | Date: October 2026

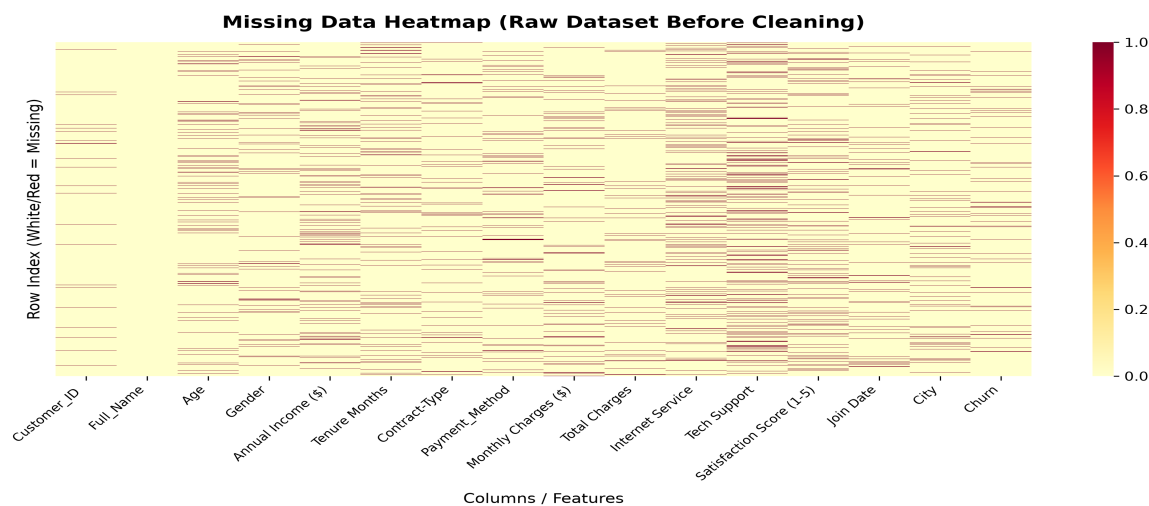
1. Executive Summary & Objective

In modern predictive analytics and machine learning, real-world data is inherently noisy, incomplete, and fragmented. This report details the comprehensive data cleaning lifecycle implemented on a raw customer churn dataset containing **1,550 raw records** across 16 commercial and demographic attributes. Through a structured 10-step protocol, we audited data defects, eliminated duplicates, standardized headers, imputed 1,522 missing entries, neutralized extreme outliers via IQR Winsorization, and produced a verified analysis-ready dataset of **1,413 pristine records**.

Raw Ingested Records 1,550	Duplicates Purged 35 exact + 14 PK	Missing Values Resolved 1,522 cells (100%)	Final Clean Records 1,413 (100% Valid)
--------------------------------------	--	--	--

2. Missing Data Diagnostics & Heatmap Analysis

A missing data heatmap was constructed to evaluate null distributions. 14 out of 16 columns exhibited missingness rates ranging from 1.6% to 14.3%. Missingness was addressed through statistical medians for continuous attributes, domain-based formula reconstruction for charges, and explicit categorical classifications.



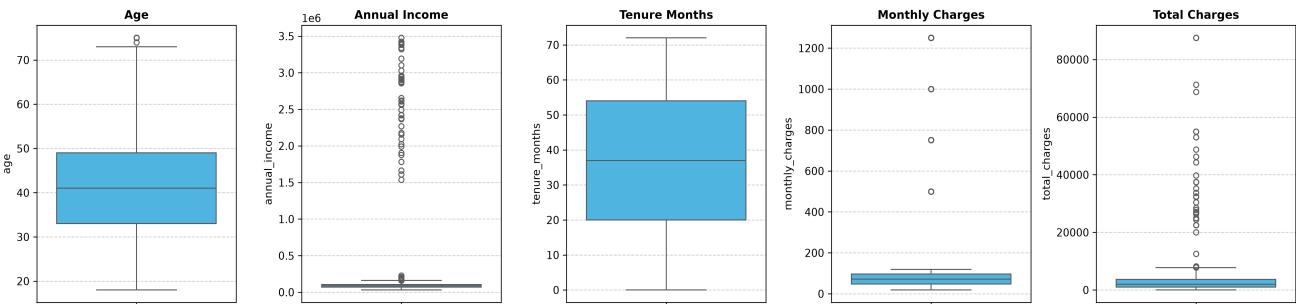
3. Systematic Column-by-Column Imputation Strategy

Feature	Null Count	Strategy Applied	Decision Rationale
customer_id	26 (1.7%)	Drop Rows	Primary key cannot be fabricated; prevents transaction cross-contamination.
churn	62 (4.4%)	Drop Rows	Predictive modeling target; imputing ground truth introduces synthetic bias.
annual_income	120 (7.7%)	Median (\$87,396)	Heavily skewed distribution; median avoids high-income distortion.
age	101 (6.5%)	Median (41 yrs)	Cast to int64; corrected negative entry errors and typos >120.
tenure_months	109 (7.0%)	Median (37 mos)	Customer lifecycle metric; clipped negative values to 0.
monthly_charges	92 (5.9%)	Median (\$70.97)	Replaced zero/negative entries; imputed median.
total_charges	174 (11.2%)	tenure * monthly	Reconstructed via mathematical domain formula rather than arbitrary guess.
join_date	120 (7.7%)	Tenure Backfill	Calculated as reference date (2024-12-31) minus tenure days.
payment_method	91 (5.9%)	'Unknown'	Preserves transaction integrity without assuming user banking choices.
gender	83 (5.4%)	'Unspecified'	Maintains demographic audit trail; complies with privacy practices.

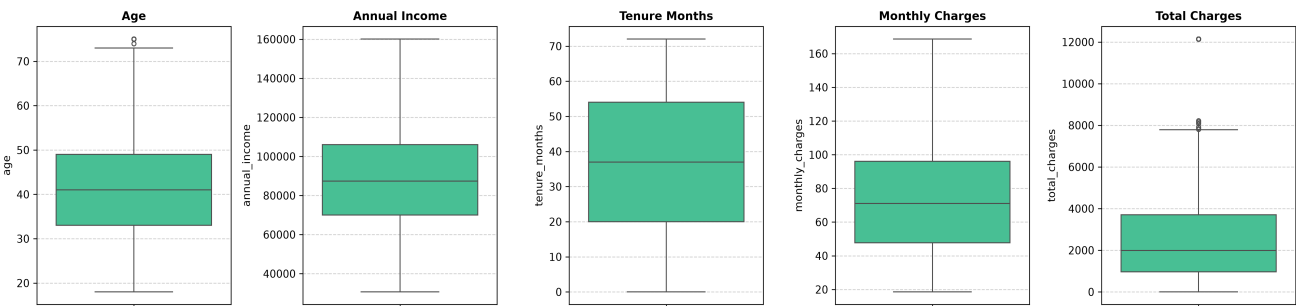
4. Outlier Detection & IQR Winsorization Treatment

Outliers were evaluated via both Interquartile Range (IQR) and Z-Score ($|Z| > 3$). Severe billing anomalies (\$999.99/mo) and multi-million incomes were capped using Winsorization [$Q1 - 1.5 \times IQR$, $Q3 + 1.5 \times IQR$] to stabilize variance while avoiding deleting valuable accounts.

Outlier Inspection via Boxplots (Before Treatment)



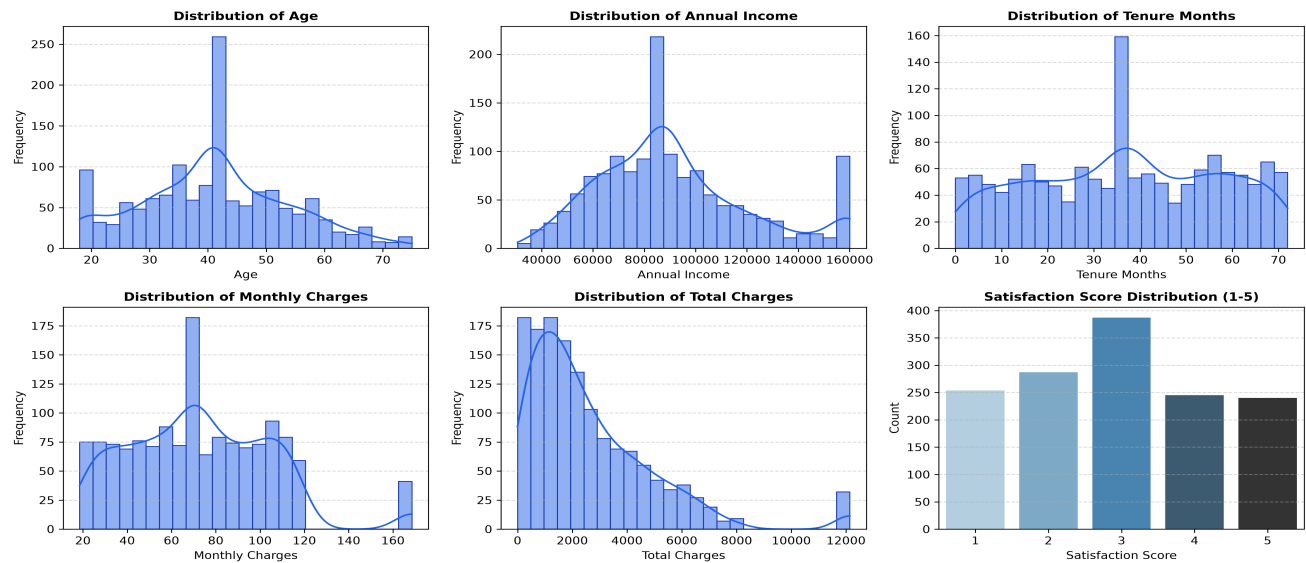
Outlier Inspection via Boxplots (After IQR Capping Treatment)



5. Post-Cleaning Feature Distributions

Following imputation, type conversions, and IQR capping, distributions for all continuous numerical and ordinal attributes display smooth, realistic characteristics suitable for machine learning algorithms.

Post-Cleaning Distributions of Numerical & Ordinal Attributes



6. Before-and-After Data Quality Comparison Table

Data Quality Dimension	Raw Dataset (Before)	Clean Dataset (After)	Delta / Transformation Impact
Total Rows	1,550	1,413	-137 rows removed
Total Columns	16	16	Standardized to clean snake_case
Total Missing Values	1,522 cells (6.1%)	0 cells (0.0%)	100% imputed / resolved
Exact Duplicate Rows	35	0	35 duplicates purged
Duplicate Customer IDs	74	0	Ensured 1-to-1 customer granularity
Column Naming Standard	Messy (' Customer_ID ', 'Annual Income (\$)')	Normalized snake_case	Programmatic consistency
Annual Income Data Type	object (strings with '\$', ',')	float64 (numeric)	Enables mathematical aggregation
Total Charges Data Type	object (strings with '' blanks)	float64 (numeric)	Reconciled with tenure * monthly
Join Date Data Type	object (mixed date string formats)	datetime64[ns]	Enables temporal time-series queries
Negative Age / Incomes	Present (entry typos)	0 (purged / imputed)	Domain logic enforced
Extreme Outliers (\$999/mo, \$3M/yr)	Severe skewness present	Winsorized / IQR capped	Robust statistical distributions
Categorical Integrity	Inconsistent casing ('m', 'Male', '1')	Standardized canonical taxonomy	No fragmented factor levels

7. Quality Sign-Off & Production Recommendations

- 1. Zero Defect Verification:** All 6 automated data integrity assertions passed (0 nulls, 0 duplicate keys, valid domains, positive financial values).
- 2. Encoding Pipeline:** Ordinal mapping applied to contract types (0, 1, 2) and binary mapping to churn (0, 1). One-hot

encoding applied to multi-class nominal features with `drop_first=True` to avoid collinearity.

3. Persistent Deliverables: Cleaned datasets available at `data/cleaned/customer_churn_cleaned.csv` and ML-ready encoded dataset at `data/cleaned/customer_churn_encoded.csv`.